

LAB Logbook

Lab Week 1

```
[1]: # Machine Learning in Finance
# Week 1
# My SID : 2546621 'Last 2 digits: 21', we'll use 21 elements

import numpy as np
```

```
[2]: array = np.arange(21)
print(array)

[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]
```

```
[3]: # Changing array to a 2D array with 1 row
array = array.reshape(1, -1)
print(array)

[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]]
```

```
[4]: # Save the original array to a different array
diff_array = array
print(diff_array)

[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]]
```

```
[5]: # shape of the diff_array
print(diff_array.shape)

(1, 21)
```

Machine Learning in Finance

Week 1

My SID : 2546621 'Last 2 digits: 21', we'll use 21 elements

import numpy as np

Making a Maxtrix with 21 Elements

array = np.arange(21)

print("Original Matrix :")

print(array)

Changing the 2D Matrix with 1 Row

array = array.reshape(1, -1)

print("\n 2D array with 1 row : ")

```
print(array)
```

```
# Saving the original array to a different array
```

```
diff_array = array
```

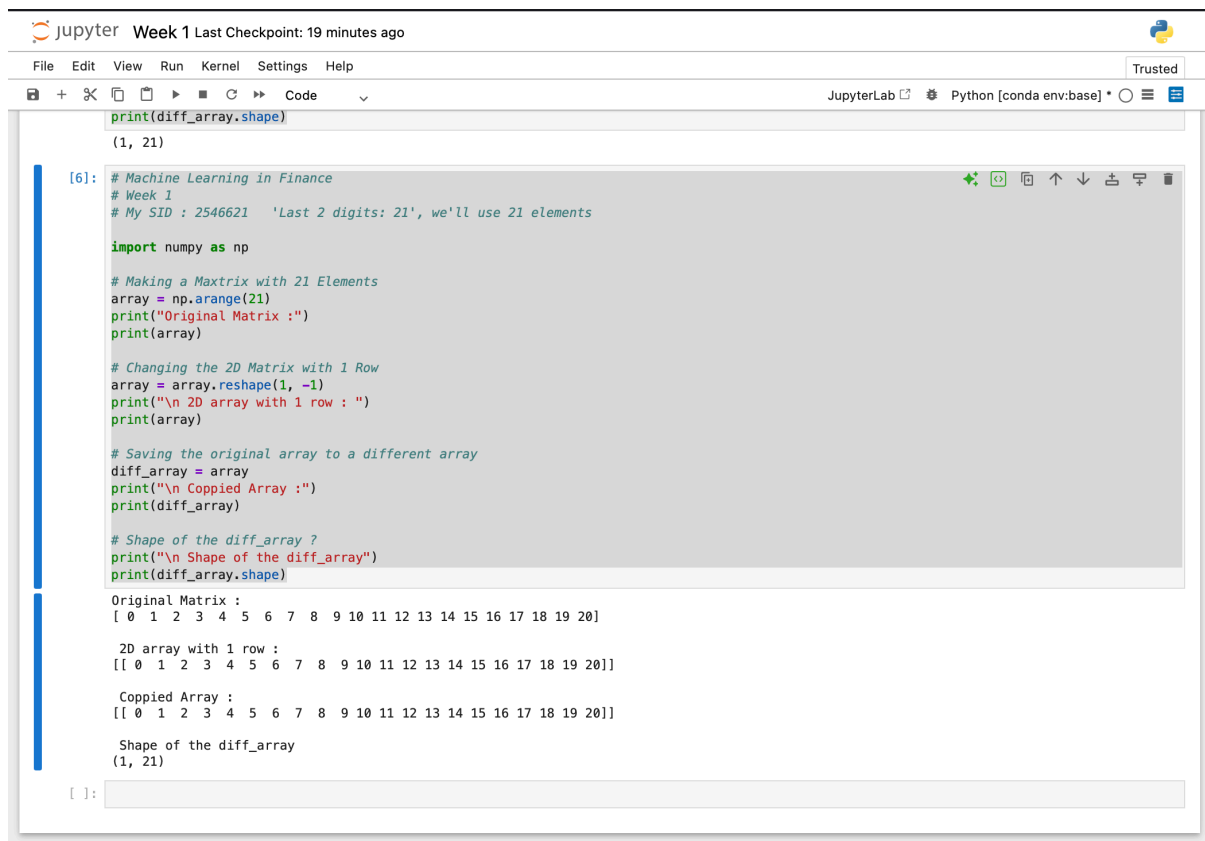
```
print("\n Coppied Array :")
```

```
print(diff_array)
```

```
# Shape of the diff_array ?
```

```
print("\n Shape of the diff_array")
```

```
print(diff_array.shape)
```



The screenshot shows a JupyterLab window titled "Week 1 Last Checkpoint: 19 minutes ago". The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and code execution. The main area contains a code editor with the following Python code:

```
print(diff_array.shape)

(1, 21)

[6]: # Machine Learning in Finance
# Week 1
# My SID : 2546621 'Last 2 digits: 21', we'll use 21 elements

import numpy as np

# Making a Maxtrix with 21 Elements
array = np.arange(21)
print("Original Matrix :")
print(array)

# Changing the 2D Matrix with 1 Row
array = array.reshape(1, -1)
print("\n 2D array with 1 row : ")
print(array)

# Saving the original array to a different array
diff_array = array
print("\n Coppied Array :")
print(diff_array)

# Shape of the diff_array ?
print("\n Shape of the diff_array")
print(diff_array.shape)
```

The output of the code is displayed below the editor:

```
Original Matrix :
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]

2D array with 1 row :
[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]]

Coppied Array :
[[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]]

Shape of the diff_array
(1, 21)
```

The output is shown in a light gray box with a vertical blue bar on the left side. The JupyterLab interface also shows a "Trusted" status in the top right corner and a "Python [conda env:base]" environment indicator.

Lab 2

Lab 3

Lab 4

Lab 5

Lab 6

Lab 7

Lab 8

Lab 9

Lab 10

Lab 11

Lab 12